

LNPTM ELCRINTM 61000iQ

DESCRIPTION

LNP ELCRIN 61000iQ, utilizing ELCRIN iQ upcycling technology, is a reinforced, impact modified PBT+PC alloy. Added features include PCR content, excellent low temperature impact, chemical resistance, colorability. This grade contains minimum 22% PCR weight content.

TYPICAL PROPERTY VALUES

Revision 20200709

PROPERTIES	TYPICAL VALUES	UNITS	TEST METHODS
MECHANICAL			
Tensile Stress, yld, Type I, 50 mm/min	45	MPa	ASTM D 638
Tensile Stress, brk, Type I, 50 mm/min	46	MPa	ASTM D 638
Tensile Strain, yld, Type I, 50 mm/min	4	%	ASTM D 638
Tensile Strain, brk, Type I, 50 mm/min	140	%	ASTM D 638
Tensile Modulus, 50 mm/min	2050	MPa	ASTM D 638
Flexural Stress, yld, 1.3 mm/min, 50 mm span	74	MPa	ASTM D 790
Flexural Modulus, 1.3 mm/min, 50 mm span	2000	MPa	ASTM D 790
Tensile Stress, yield, 50 mm/min	49	MPa	ISO 527
Tensile Stress, break, 50 mm/min	47	MPa	ISO 527
Tensile Strain, yield, 50 mm/min	4	%	ISO 527
Tensile Strain, break, 50 mm/min	150	%	ISO 527
Tensile Modulus, 1 mm/min	1950	MPa	ISO 527
Flexural Stress, yield, 2 mm/min	70	MPa	ISO 178
Flexural Modulus, 2 mm/min	1920	MPa	ISO 178
IMPACT			
Izod Impact, notched, 23°C	650	J/m	ASTM D 256
Izod Impact, notched, -30°C	600	J/m	ASTM D 256
Instrumented Impact Total Energy, 23°C	50	J	ASTM D 3763
Izod Impact, notched 80*10*4 +23°C	55	kJ/m ²	ISO 180/1A
Izod Impact, notched 80*10*4 -30°C	47	kJ/m ²	ISO 180/1A
Charpy 23°C, V-notch Edgew 80*10*4 sp=62mm	55	kJ/m ²	ISO 179/1eA
THERMAL			
HDT, 1.82 MPa, 3.2mm, unannealed	75	°C	ASTM D 648
HDT, 0.45 MPa, 3.2 mm, unannealed	99	°C	ASTM D 648
CTE, -40°C to 150°C, flow	9E-05	1/°C	ASTM E 831
CTE, -40°C to 150°C, xflow	9.5E-05	1/°C	ASTM E 831
CTE, -30°C to 80°C, flow	9.E-05	1/°C	ISO 11359-2
CTE, -30°C to 80°C, xflow	9.5E-05	1/°C	ISO 11359-2
Relative Temp Index, Elec ⁽¹⁾	75	°C	UL 746B
Relative Temp Index, Mech w/impact ⁽¹⁾	75	°C	UL 746B
Relative Temp Index, Mech w/o impact ⁽¹⁾	75	°C	UL 746B
PHYSICAL			
Specific Gravity	1.19	-	ASTM D 792
Mold Shrinkage, flow, 3.2 mm	0.8 – 1.2	%	SABIC method
Melt Flow Rate, 250°C/5.0 kgf	13	g/10 min	ASTM D 1238

PROPERTIES	TYPICAL VALUES	UNITS	TEST METHODS
Density	1.2	g/cm ³	ISO 1183
Water Absorption, (23°C/saturated)	0.3	%	ISO 62-1
Moisture Absorption (23°C / 50% RH)	0.08	%	ISO 62
Melt Volume Rate, MVR at 250°C/5.0 kg	12	cm ³ /10 min	ISO 1133
FLAME CHARACTERISTICS ⁽¹⁾			
UL Yellow Card Link	E207780-100803460	-	-
UL Recognized, 94HB Flame Class Rating	≥1.5	mm	UL 94
INJECTION MOLDING			
Drying Temperature	110	°C	
Drying Time	4 – 6	hrs	
Drying Time (Cumulative)	8	hrs	
Maximum Moisture Content	0.02	%	
Melt Temperature	255 – 275	°C	
Nozzle Temperature	250 – 265	°C	
Front - Zone 3 Temperature	250 – 270	°C	
Middle - Zone 2 Temperature	245 – 265	°C	
Rear - Zone 1 Temperature	240 – 260	°C	
Mold Temperature	40 – 90	°C	
Back Pressure	0.3 – 0.7	MPa	
Screw Speed	40 – 80	rpm	
Shot to Cylinder Size	50 – 80	%	
Vent Depth	0.013 – 0.02	mm	

(1) UL Ratings shown on the technical datasheet might not cover the full range of thicknesses and colors. For details, please see the UL Yellow Card

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